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SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS & SCIENCE

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Self Financed B.B.A.-B.Com. (EM & GM)-B.C.A. (Regular/AI), B.Sc. (DS), M.Sc. (IT), M.Com. (EM & GM) Degree Programme

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Bachelor Of Computer Applications (BCA)

Project Report

BCA Sem VI

AY 2025-26

Project Title : Health And Fitness Club

SUBMITTED BY:

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SUBMITTED TO:

MS. DAMYANTI K. PATEL

ACKNOWLEDGEMENT

We take this opportunity to express our deep sense of gratitude to all those who helped us successfully complete our project titled **"FitSoul – Health and Fitness Club Website."** This project has been an enriching learning experience, and it would not have been possible without the guidance, support, and encouragement of several individuals.

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We take this moment to appreciate the support and encouragement of our **friends and classmates**, who consistently helped us with ideas, suggestions, and feedback throughout the project development process. Their inspiration and collaboration kept us motivated during challenging stages.

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With heartfelt gratitude, we thank everyone who directly or indirectly contributed to the success of this project.

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1. INTRODUCTION

1.1 Project Description

The **FitSoul Health and Fitness Club Website** is an innovative web-based platform developed to enhance the way individuals engage with fitness services and resources. In today's fast-paced lifestyle, people are increasingly conscious about their health, and digital platforms have become a primary source for fitness-related guidance. FitSoul addresses this need by providing a centralized hub where users can access information about workouts, nutrition, memberships, and expert training sessions.

The primary goal of the website is to create an interactive environment that motivates users to adopt a healthy lifestyle. Visitors can explore different fitness programs, learn about the club's facilities, and get personalized recommendations that suit their fitness levels. For members, the website provides convenient access to class schedules, booking systems, and exclusive fitness resources. This helps streamline communication between the fitness club and its community while promoting active engagement.

Technologically, FitSoul is designed with a responsive and modern interface, ensuring accessibility on desktops, tablets, and mobile devices. The user-friendly design enhances the browsing experience, allowing users to navigate effortlessly through workout plans, health tips, and service details. Moreover, the integration of multimedia elements such as images, videos, and testimonials adds to the website's appeal and credibility.

Beyond fitness guidance, the website also focuses on building a supportive fitness community. It allows users to stay updated with upcoming events, workshops, and promotional offers. By connecting individuals with expert trainers and like-minded peers, FitSoul encourages users to stay committed to their fitness journey.

In conclusion, the FitSoul Health and Fitness Club Website is not just an informational portal but a complete fitness companion. It bridges the gap between the fitness club and its members, offering convenience, motivation, and professional support under one platform. The project plays a vital role in promoting a healthier lifestyle while expanding the digital presence of the fitness club.

❖ Admin Panel

✓ It is mandatory to login with an admin account.

✓ Admin can manage all user accounts including members.

✓ Admin has full control to add, update, or delete fitness programs, Membership plans.

✓ Admin can upload and manage images, and other content displayed on the website.

✓ Admin can view membership details, payment records, and overall reports of the fitness club.

✓ Admin can send notifications and important announcements to all users (e.g., holiday notice, new offers, health tips).

✓ Admin can monitor feedback/messages sent by users and respond accordingly.

❖ User Panel

✓ It is mandatory to login/register with a user account to access member features.

✓ Users can update their personal profile including health details and fitness goals.

✓ Users can browse available fitness programs, diet tips, and trainer information.

✓ Users can check class schedules, book training sessions, and manage their memberships online.

✓ Users can view their payment history, membership status, and upcoming renewal dates.

✓ Users can communicate with the admin or trainers through a built-in messaging/contact system.

✓ Users can stay updated with notices such as new events, discounts, and health awareness campaigns.

✓ New users can sign up for memberships directly through the website and request admin approval if required.

1.2 Project Profile

Project Title	FITSOUL
Defination	The FITSOUL is an innovative web-based platform developed to enhance the way individuals engage with fitness services and resources.
Devloper For	Sutex Bank Collage Of Computer Application & Science
Fronted	Angular.JS
Backend	Node.JS , Express.JS
Operating System	Microsoft window 11
Programming Language	Web
Project Guide	Damyanti K. Patel

2. Environment Description

2. Environment Description

The development and deployment of the **FitSoul Health and Fitness Club Website** requires a specific computing environment. This environment consists of both **hardware** and **software components** that ensure smooth operation during the stages of development, testing, and final implementation. The selection of these components is based on cost-effectiveness, performance, scalability, and ease of use.

2.1 Hardware Requirements

To design, develop, and run the FitSoul website, the following hardware configurations are recommended:

Minimum Requirements (for Development & Testing)

- **Processor (CPU):** Intel Core i3 / AMD Ryzen 3 or above
- **RAM:** 4 GB
- **Storage:** 256 GB HDD / SSD
- **Display:** 1024x768 resolution monitor
- **Network:** Stable internet connection (minimum 5 Mbps)

Recommended Requirements (for Smooth Performance)

- **Processor (CPU):** Intel Core i5/i7 or AMD Ryzen 5/7 (Quad Core, 2.5 GHz or higher)
- **RAM:** 8 GB or higher (to support Node.js, MongoDB, and Angular simultaneously)
- **Storage:** 512 GB SSD (faster data access and smooth IDE performance)
- **Display:** Full HD Monitor (1920x1080) for better design and testing visibility
- **Peripherals:** Keyboard, Mouse, Webcam (for trainer/user interactions, if needed)
- **Network:** High-speed broadband internet (minimum 25 Mbps for testing APIs, video uploads)

2.2 Software Requirements

The following software tools and platforms are essential for project development, testing, and deployment:

Operating System

- Development: Microsoft Windows 10/11 or Linux (Ubuntu recommended)
- Deployment: Linux (Ubuntu Server / CentOS) for better server stability

Frontend Development

- **Angular** (UI framework for building dynamic and responsive web applications)
- **Angular Router** (for navigation between pages)

Backend Development

- **Node.js** (JavaScript runtime environment)
- **Express.js** (backend framework for RESTful APIs)

Database

- **MongoDB** (NoSQL database for scalability and flexible data storage)

Development Tools

- **Visual Studio Code** (code editor with extensions for Angular/Node)
- **Postman** (API testing tool)
- **MongoDB Compass** (GUI for database management)
- **Git & GitHub** (version control and collaboration)

Browser Support (for Testing)

- Google Chrome (latest version)
- Mozilla Firefox
- Microsoft Edge
- Safari

Other Software

- **Draw.io / SmartDraw / Lucidchart** – for creating diagrams (DFD, ERD, UML, etc.)
- **Figma / Adobe XD** – for UI/UX design and wireframes
- **MS Word / Google Docs** – for documentation
- **MS PowerPoint / Google Slides** – for project presentation

2.3 Technologies Used

The FitSoul project integrates modern web technologies for efficiency, scalability, and user-friendliness:

1. **Frontend Technology**:
    - **Angular.js** UI framework for building dynamic and responsive web applications
    - **CSS Frameworks** like Bootstrap/Tailwind CSS for responsive design.
2. **Backend Technology**:
    - **Node.js** for server-side scripting.
    - **Express.js** for RESTful APIs and routing.
3. **Database**:
    - **MongoDB** (NoSQL) for storing user details, memberships, bookings, and payments.
4. **Authentication & Security**:
    - **JWT (JSON Web Token)** for secure user authentication.
    - **bcrypt.js** for password hashing.
5. **Version Control & Collaboration**:
    - **GitHub** for team collaboration and code versioning.
6. **Deployment & Hosting**:
    - **Heroku / AWS / Netlify / Vercel** for hosting web application.
    - **MongoDB Atlas** for cloud database management.

3. System Analysis and Planning

System Analysis is the process of understanding the current system, identifying its drawbacks, and planning an improved solution. For the **FitSoul Health and Fitness Club Website**, system analysis helps define the project's scope, feasibility, and requirements. It ensures that the developed system meets both **functional** and **non-functional** expectations of users, admins, and trainers.

3.1 Existing System and its Drawbacks

Existing System

Currently, most fitness clubs operate either:

- **Manual Systems**: Using registers, paper forms, or spreadsheets to manage memberships, bookings, and payments.
- **Basic Websites**: Some fitness centers may have static websites that provide only contact details and general information without online interaction.
- **Third-party Apps**: Some rely on external apps for class bookings and payments, but they lack personalization for the specific fitness club.

Drawbacks of Existing System

1. **Manual Errors**
   - Mistakes in membership renewals, payments, and class schedules due to human error.
2. **Time-Consuming**
   - Trainers and admins spend a lot of time handling member queries manually.
3. **Limited Accessibility**
   - Members cannot access real-time information about schedules, bookings, or membership status.
4. **Lack of Automation**
   - No automatic notifications for payments, renewals, or new events.
5. **Poor User Engagement**
   - No personalized recommendations, fitness tips, or interactive content for members.

6. **Data Management Issues**
    - Difficult to maintain user profiles, payment records, and health details in manual or semi-digital systems.
7. **Security Concerns**
    - Data stored in registers or basic spreadsheets is vulnerable to misuse or loss.

3.2 Feasibility Study

The feasibility study examines whether the proposed system is **practical, economical, and technically achievable**.

1. Technical Feasibility

- **Frontend**: Angular.js provides a dynamic, responsive, and user-friendly interface.
- **Backend**: Node.js with Express.js ensures fast server-side processing.
- **Database**: MongoDB provides scalable, flexible, and secure data storage.
- **Hosting**: Deployment can be done on Heroku, Vercel, or AWS, ensuring scalability.

2. Economic Feasibility

- Development tools like **VS Code, GitHub, Node.js, and MongoDB** are free or open-source.
- Hosting services like **Netlify/Vercel (Frontend)** and **MongoDB Atlas (Database)** offer free tiers.
- The cost of development is low, requiring only basic hardware and internet.

3. Operational Feasibility

- The system provides an **easy-to-use interface** for both admin and users.
- Reduces workload of trainers and staff by automating bookings, memberships, and payments.
- Increases customer satisfaction through real-time updates and digital access.

4. Schedule Feasibility

- The system can be developed within **4–6 months**, divided into:
    - Requirement Gathering – 1 weeks
    - Design Phase – 1 weeks

- Development – 2 weeks
- Testing – 2 weeks
- Deployment & Documentation – 1 weeks

3.3 Requirement Gathering and Analysis

Requirement gathering involves identifying what **users, admins, and trainers** need from the system. These requirements are categorized into **Functional** and **Non-Functional** requirements.

A. Functional Requirements

Admin Panel

- Login with admin credentials.
- Manage user accounts (add, update, delete members).
- Add, update, and delete fitness programs and membership plans.
- Manage payments, class schedules, and trainer assignments.
- Send notifications, health tips, and event updates.
- View feedback/messages from users and reply.
- Generate reports for memberships and payments.

User Panel

- Register and login securely.
- Update personal details, health information, and fitness goals.
- Browse fitness programs, trainers, and diet tips.
- Book classes or training sessions online.
- Manage memberships (purchase, renew, cancel).
- View payment history and upcoming renewals.
- Send feedback or queries to admin/trainers.
- Get notifications about events, offers, and updates.

Trainer Panel (Optional Feature)

- View assigned classes and users.
- Update availability schedules.
- Share workout or diet plans with users.
- Interact with users via built-in messaging.

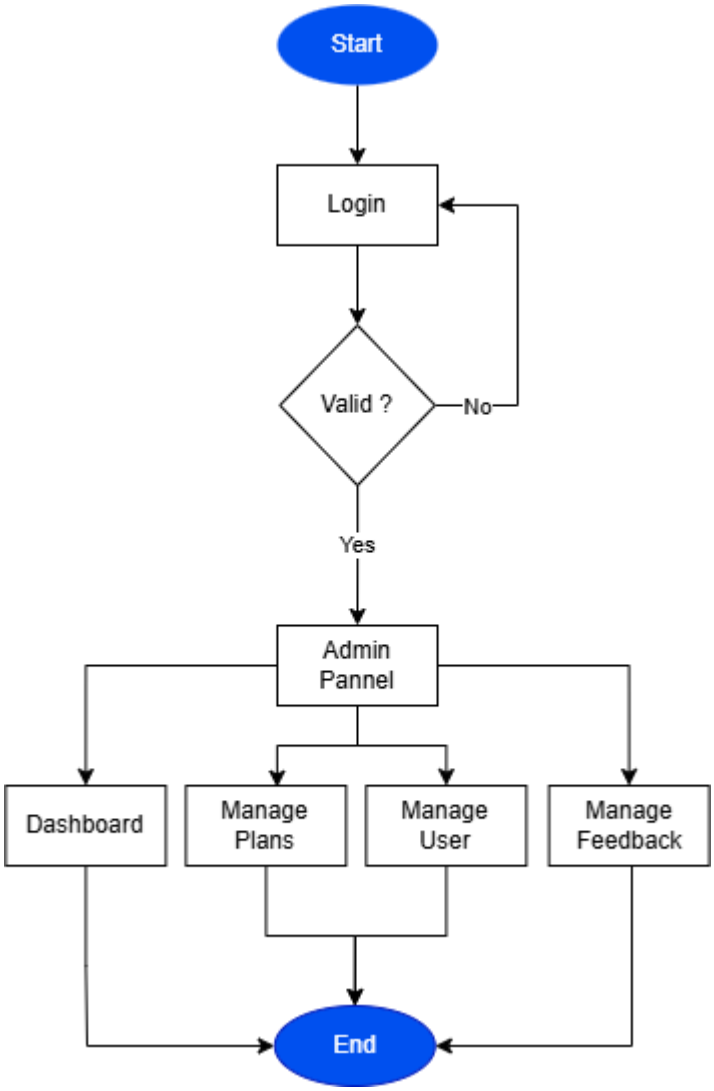
B. Non-Functional Requirements

1. **Performance**
    - The system should load within **3 seconds** under normal conditions.
    - Must support at least **100 concurrent users**.
2. **Security**
    - User passwords stored with **bcrypt encryption**.
    - Authentication with **JWT tokens**.
    - Role-based access (Admin/User/Trainer).
3. **Usability**
    - Responsive design for desktop, tablet, and mobile.
    - Simple, user-friendly navigation.
4. **Reliability & Availability**
    - 99.9% uptime when hosted on cloud servers.
    - Auto backup for databases.
5. **Scalability**
    - Can expand to support multiple branches of fitness clubs.
    - Easily integrate with third-party payment gateways (Razorpay/Stripe/PayPal).
6. **Maintainability**
    - Modular code using Angular components and Node.js services.
    - Proper documentation for future upgrades.

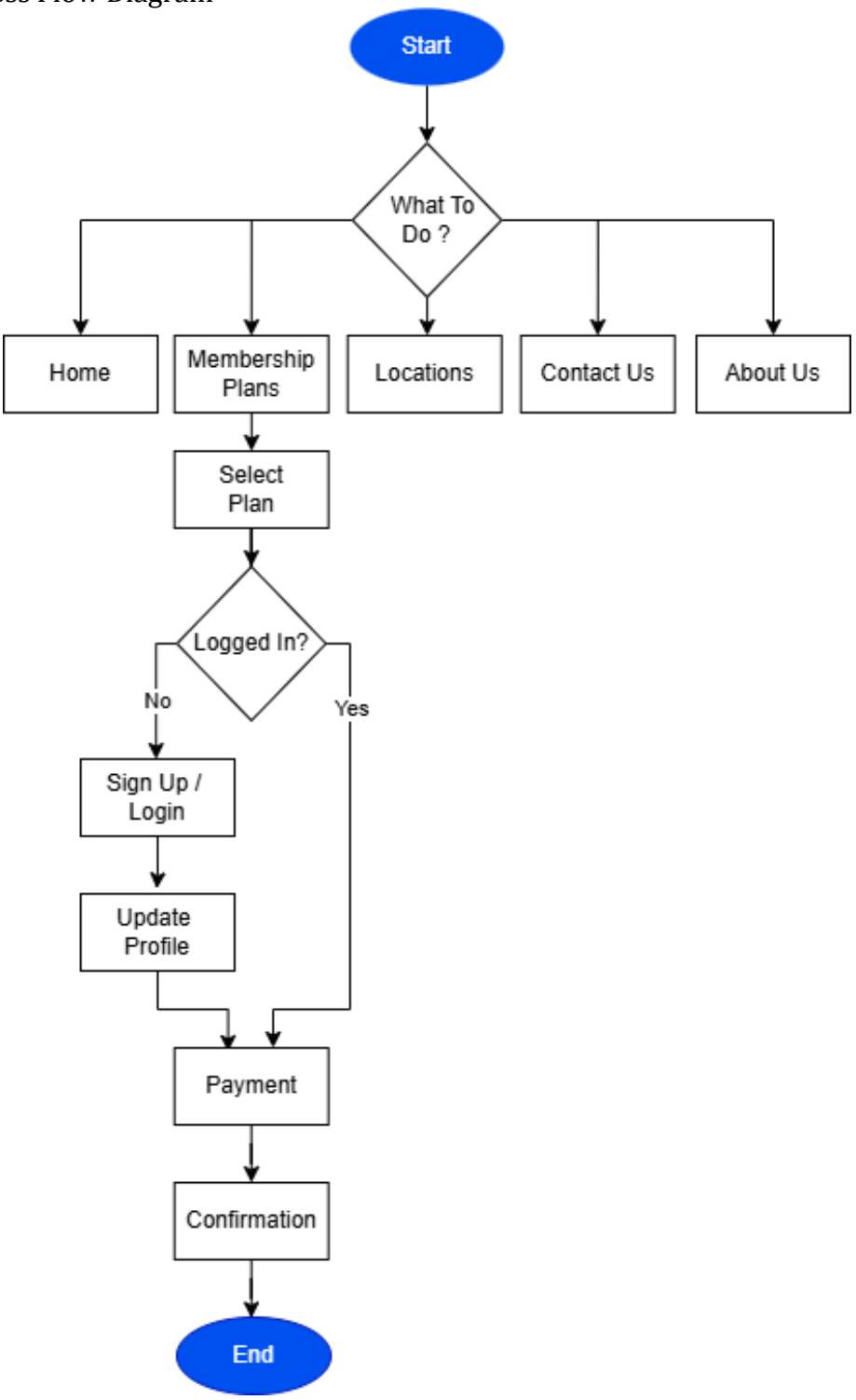
4. Detail Planning

4.1 System Flow Diagram

- Admin Process Flow Diagram

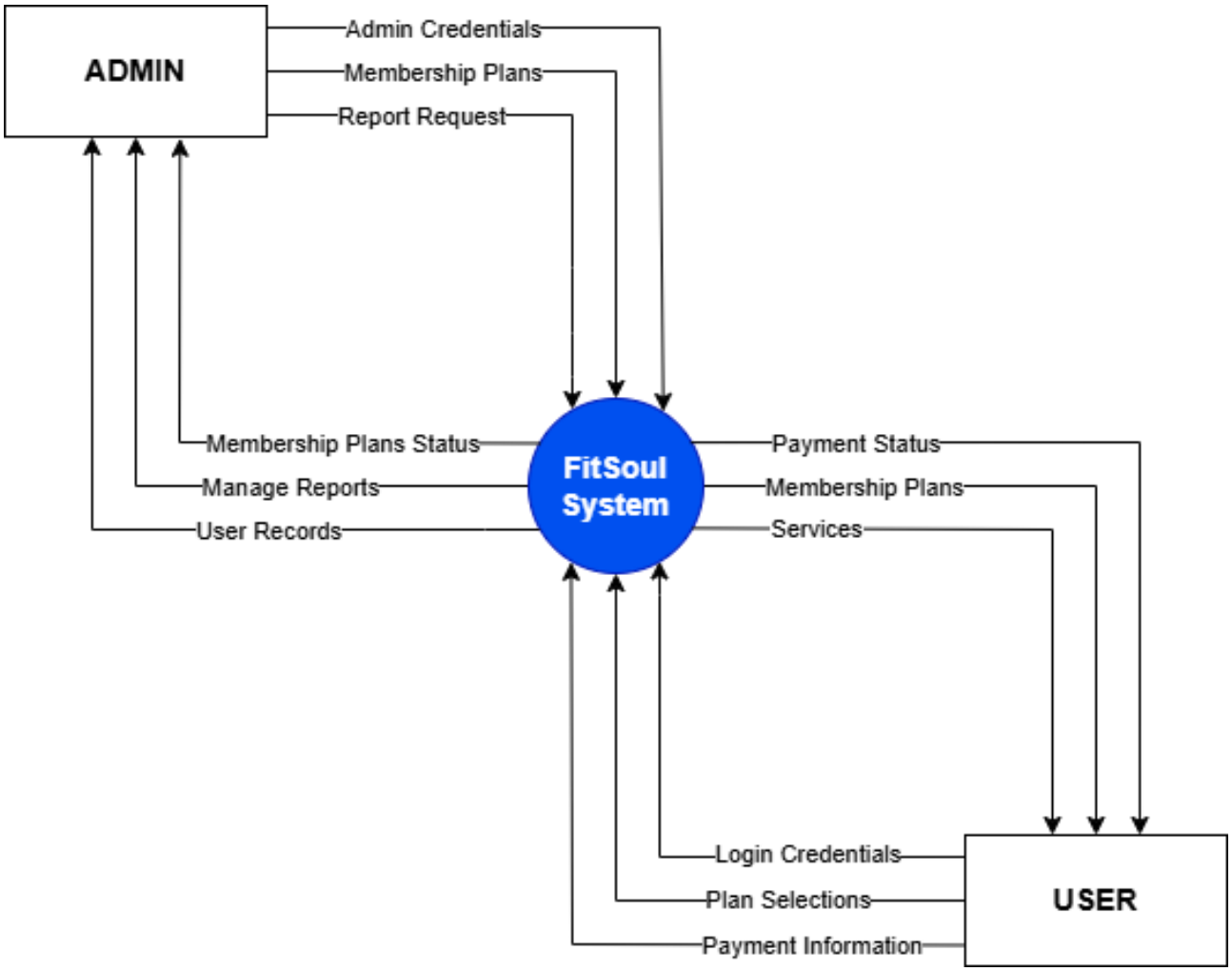


- User Process Flow Diagram

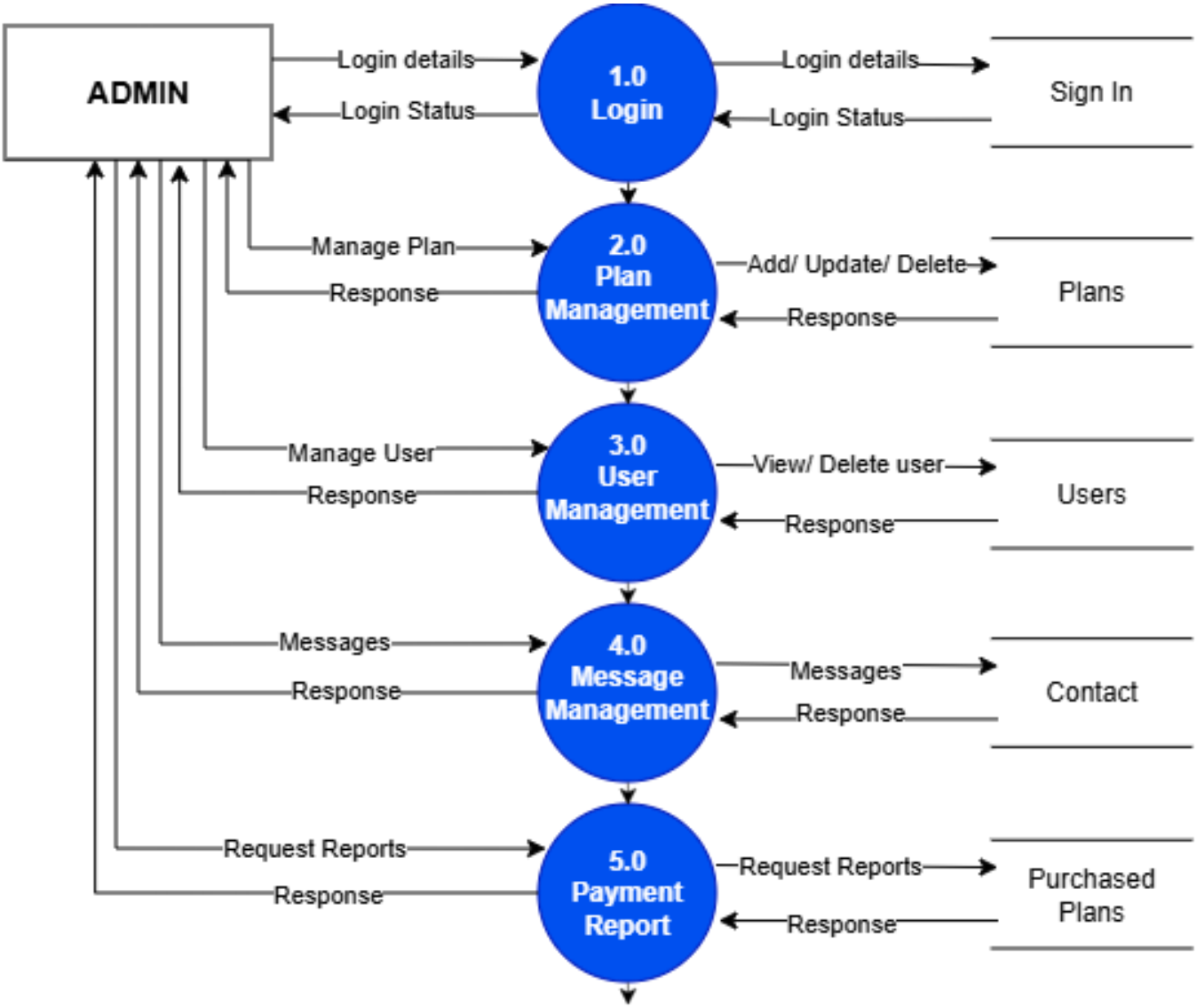


4.2 Data Flow Diagram / UML

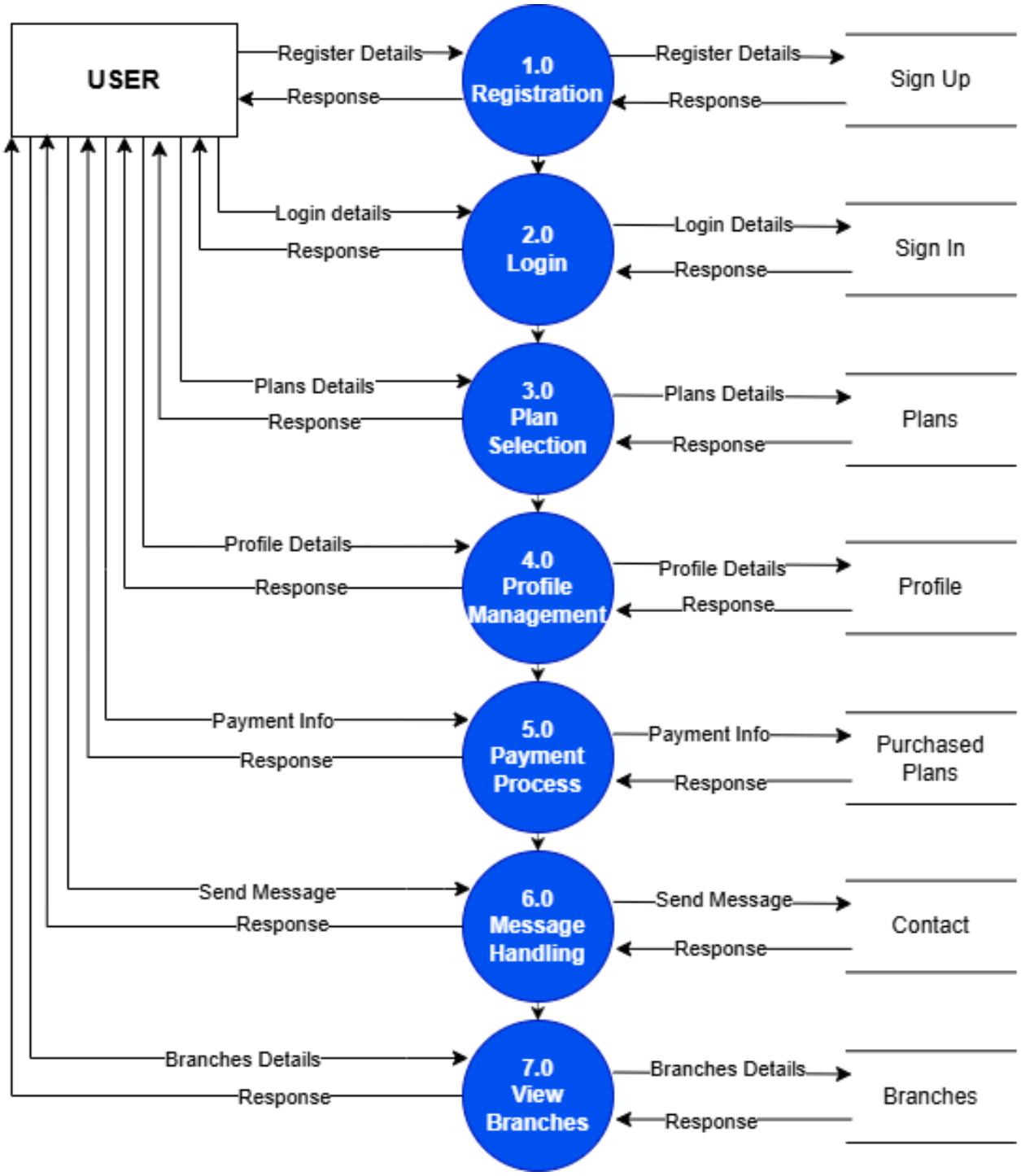
-: 0 Level Diagram :-



-: 1st Level Admin Diagram :-

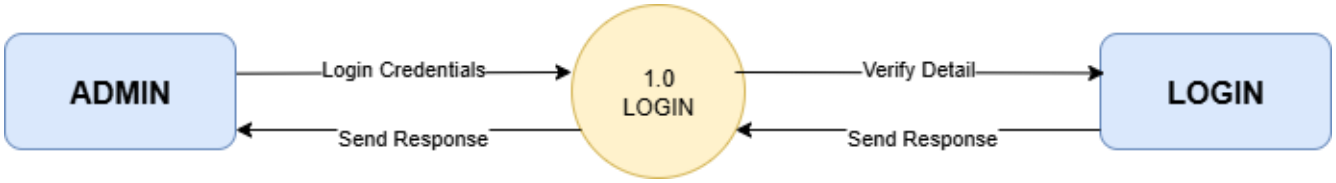


-: 1st Level User Diagram :-

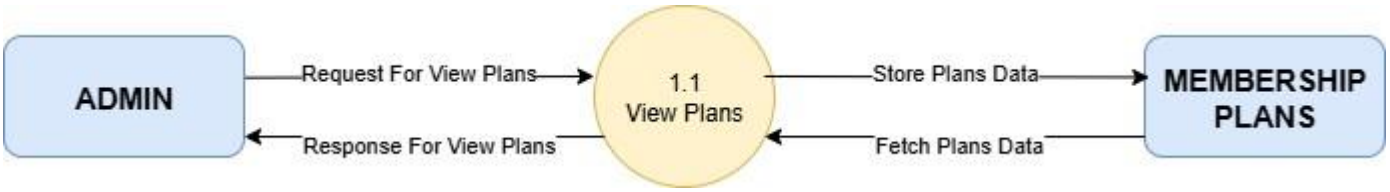


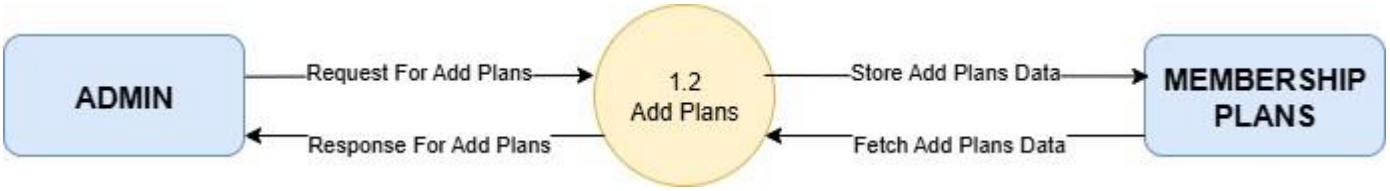
-: 2nd Level Admin Diagram :-

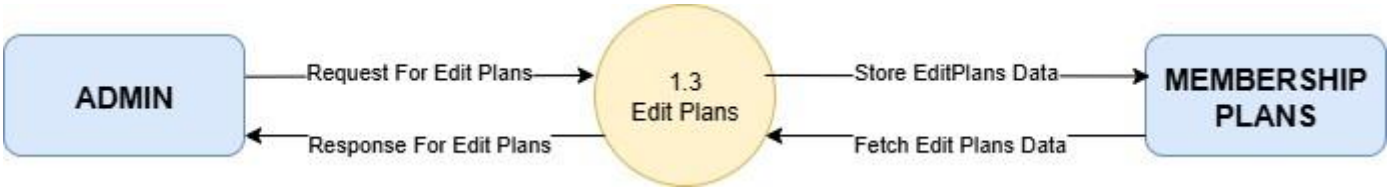
❖ Login

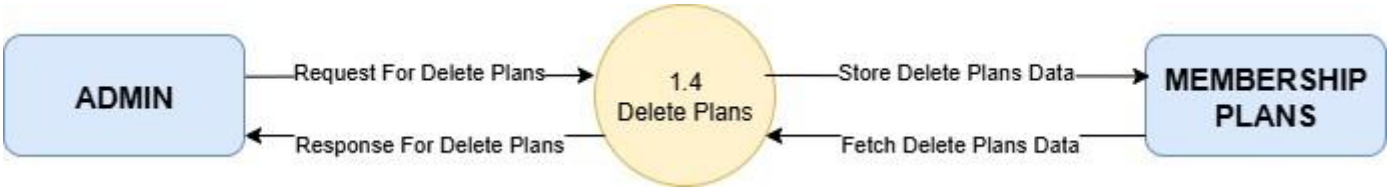


❖ Manage Plans

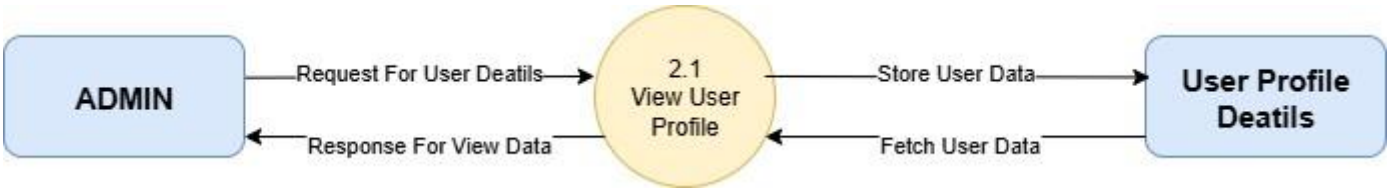




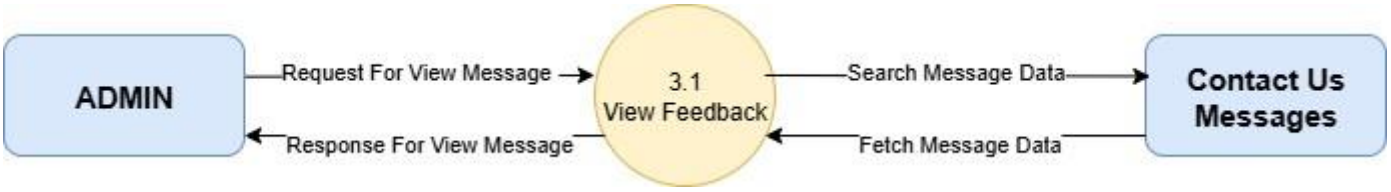




- **Manage Users**

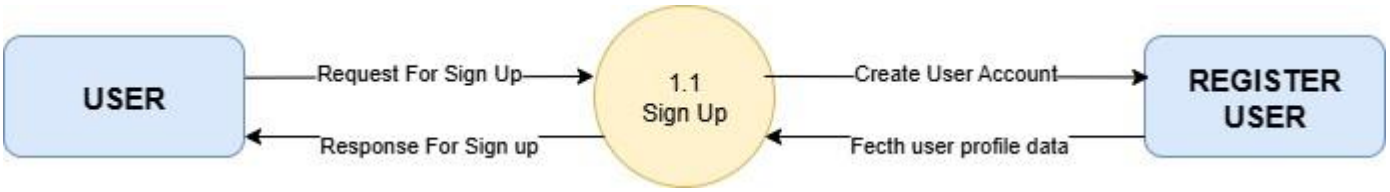


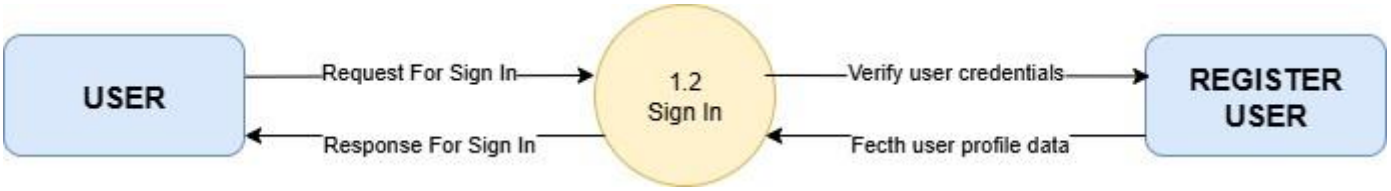
- **User Feedback**



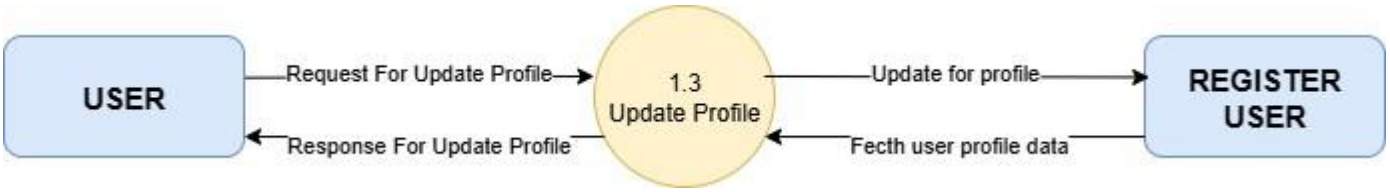
-: 2nd Level User Diagram :-

- Login

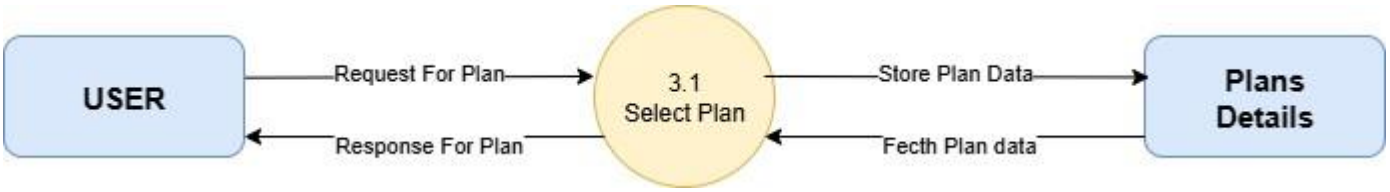


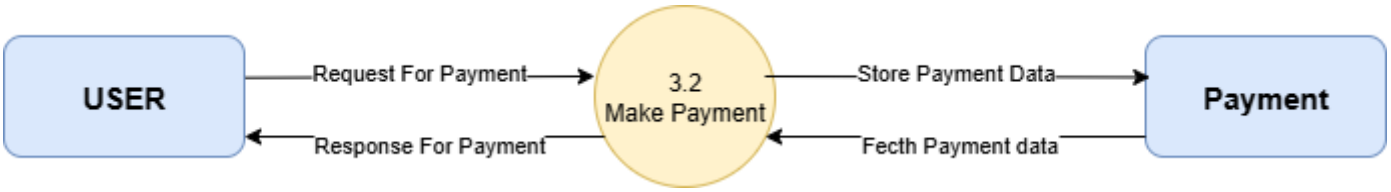


- Update Profile

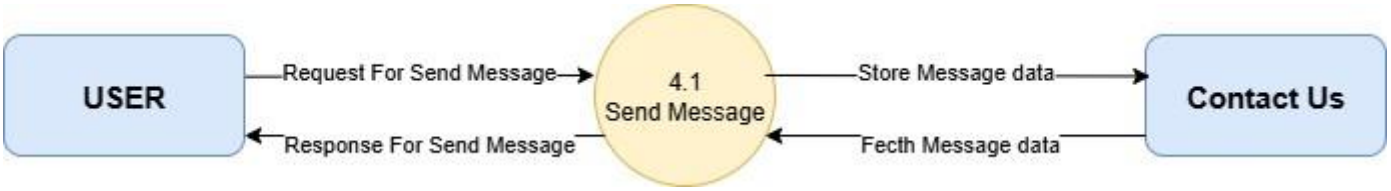


- Membership Plans

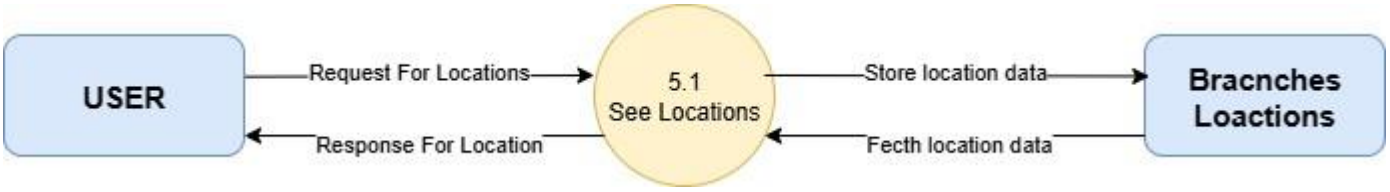




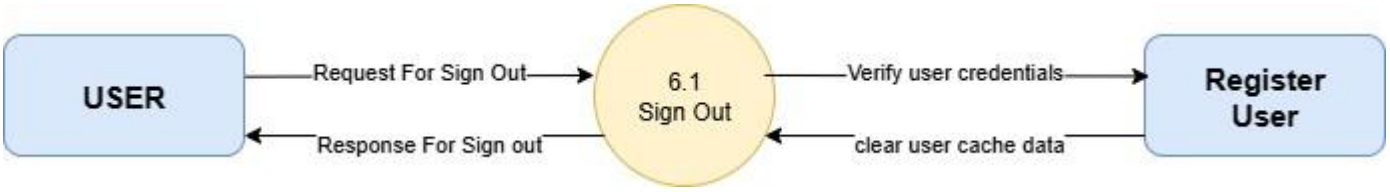
- Send Message



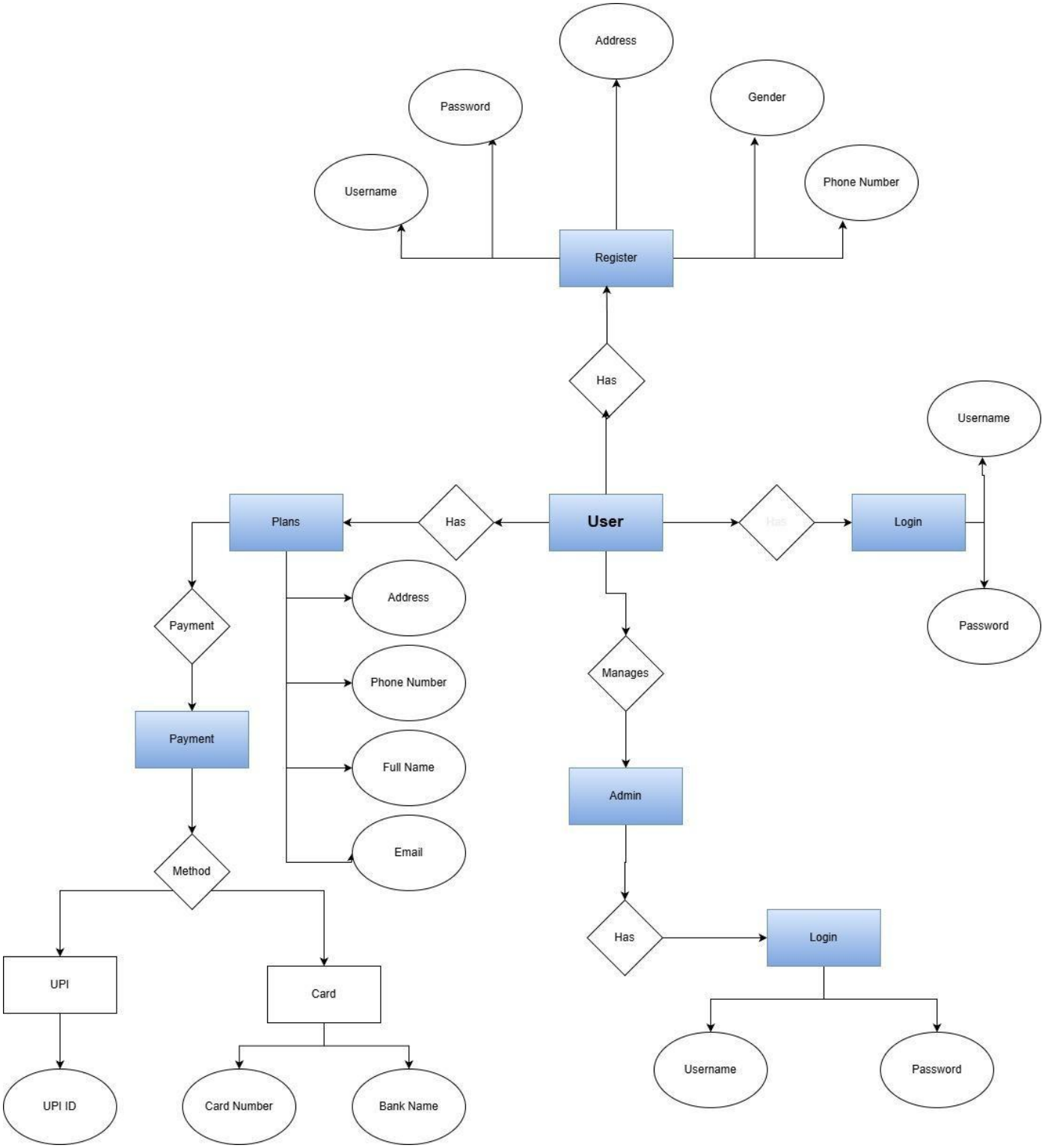
- Branches Locations



- Log Out



4.3 Entity-Relationship Diagram



4.4 Data Dictionary

Admin

Field Name	Data Type	Constraint	Description
admin_id	INT	Primary Key	Unique identifier for the Admin.
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username for the Admin Panel.
password_hash	VARCHAR(255)	NOT NULL	Hashed and salted password for security.
email	VARCHAR(100)	UNIQUE, NOT NULL	Admin contact email.
full_name	VARCHAR(100)	NOT NULL	Full name of the admin.
last_login	DATETIME	NULL	Timestamp of the last successful login.
role	VARCHAR(20)	NOT NULL	E.g., 'SuperAdmin', 'ContentManager'.

User

Field Name	Data Type	Constraint	Description
user_id	INT	Primary Key, NOT NULL	Unique identifier for the user/member.
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
mobile_no	VARCHAR(15)	UNIQUE, NOT NULL	User's contact phone number.
gender	VARCHAR(10)	NOT NULL	User's gender
address	VARCHAR(255)	NULL	residential address.
email	VARCHAR(100)	UNIQUE, NOT NULL	User's contact email.
password	VARCHAR(255)	NOT NULL	password for security.
registration_date	DATETIME	NOT NULL	Date and time the account was created.
current_plan_id	INT	Foreign Key (Plans)	ID of the user's currently active membership plan

- Membership Plans

Field Name	Data Type	Constraint	Description
payment_id	INT	Primary Key	Unique identifier for the payment transaction.
user_id	INT	Foreign Key (Users)	The user who made the payment.
plan_id	INT	Foreign Key (Plans)	The membership plan purchased.
amount_paid	DECIMAL(10, 2)	NOT NULL	The exact amount of the transaction.
transaction_date	DATETIME	NOT NULL	Date and time the payment was processed.
payment_method	VARCHAR(50)	NULL	E.g., 'Credit Card', 'UPI', 'Cash'.
status	VARCHAR(20)	NOT NULL	E.g., 'Completed', 'Pending', 'Failed'.

5. REFERENCE

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